

DFT (Design for Testability)

Parameter	Value	Status
Design	8-bit ALU	-
Total Flip-Flops	16	✓
Scan Flip-Flops Inserted	16	✓
Number of Scan Chains	1	✓
Scan Chain Length	16	✓
Scan Enable Signal	scan_en	✓
Scan Input	scan_in	✓
Scan Output	scan_out	✓
Scan Clock	clk	✓
Stuck-at Fault Model	Single Stuck-at (SA0/SA1)	✓
Estimated Fault Coverage	98.2%	PASS
ATPG Status	Completed	PASS

DFT Tool Log :-

DFT Scan Insertion Report

Design Name : alu

Scan Style : Multiplexed Scan

Total Flip-Flops : 16

Scan Flip-Flops : 16

Scan Chains : 1

Scan Chain Length : 16

Scan Enable : scan_en

Scan Input : scan_in

Scan Output : scan_out

Estimated Stuck-at Coverage : 98.2 %

Unscannable Registers : 0

DFT Rule Violations : 0

Status : PASS

LVS Log :-

Layout Versus Schematic Report

Devices Compared : 52

Devices Matched : 52

Nets Compared : 68

Nets Matched : 68

Open Nets : 0

Shorted Nets : 0

Property Mismatch : 0

Final LVS Result : CLEAN

LVS Report

Check	Result	Status
Layout Netlist Generated	Yes	PASS
Schematic Netlist Generated	Yes	PASS
Devices Matched	52/52	PASS
Nets Matched	68/68	PASS

Check	Result	Status
Shorts	0	PASS
Opens	0	PASS
Parameter Mismatch	0	PASS
LVS Result	CLEAN	PASS

DRC Log

Design Rule Check Report

Minimum Width	: PASS
Minimum Spacing	: PASS
Metal Spacing	: PASS
Via Enclosure	: PASS
Poly Enclosure	: PASS
Well Spacing	: PASS
Antenna Rule	: PASS
Total Violations	: 0
DRC Status	: CLEAN

DRC Report

Rule	Violations	Status
Minimum Width	0	PASS
Minimum Spacing	0	PASS
Metal Overlap	0	PASS
Via Enclosure	0	PASS

Rule	Violations	Status
Poly Spacing	0	PASS
Well Spacing	0	PASS
Antenna Violations	0	PASS
Total Violations	0	PASS

Stuck-at Fault Coverage

Fault Type	Total	Detected	Coverage
Stuck-at-0	620	608	98.1%
Stuck-at-1	620	610	98.4%
Overall	1240	1218	98.2%